

Resultatkörning, screenshots

Modell träningsdata 2026-06-28

```
=====
      TRAINING SUMMARY
=====
```

	group	model	score
0	G1	RFModel	0.755985
1	G1	HGBModel	0.954440
2	G1	CatBoostModel	0.946718
3	G2	RFModel	0.731757
4	G2	HGBModel	0.960135
5	G2	CatBoostModel	0.945270
6	G3	RFModel	0.708108
7	G3	HGBModel	0.959073
8	G3	CatBoostModel	0.945174
9	G4	RFModel	0.769178
10	G4	HGBModel	0.969178
11	G4	CatBoostModel	0.956849

Optimering av modellerna med hjälp av Optuna.

```
=== OPTIMIZATION SUMMARY ===
```

	group	model	best_score	best_params
0	G1	RFModel	0.938996	{'n_estimators': 698, 'max_depth': 10, 'min_sa...
1	G1	HGBModel	0.944402	{'learning_rate': 0.06274227849643471, 'max_de...
2	G1	CatBoostModel	0.949807	{'depth': 5, 'learning_rate': 0.17459145084880...
3	G2	RFModel	0.954730	{'n_estimators': 895, 'max_depth': 12, 'min_sa...
4	G2	HGBModel	0.945946	{'learning_rate': 0.05232761314412046, 'max_de...
5	G2	CatBoostModel	0.952027	{'depth': 6, 'learning_rate': 0.05004228961223...
6	G3	RFModel	0.961390	{'n_estimators': 728, 'max_depth': 12, 'min_sa...
7	G3	HGBModel	0.957529	{'learning_rate': 0.04172774325253595, 'max_de...
8	G3	CatBoostModel	0.952124	{'depth': 6, 'learning_rate': 0.03590671001797...
9	G4	RFModel	0.969178	{'n_estimators': 760, 'max_depth': 10, 'min_sa...
10	G4	HGBModel	0.965753	{'learning_rate': 0.038803551706789785, 'max_d...
11	G4	CatBoostModel	0.967808	{'depth': 5, 'learning_rate': 0.03951010897194...

En retraining (omträning) genomfördes där modellerna tränades om på train+val-datan med de optimerade hyperparametrarna.

```
=====
STARTING FULL RETRAIN PIPELINE
=====

=====
RETRAINING MODELS ON G1
=====

--- Retraining RFModel on G1 ---
Saved RFModel → c:\Users\Eier\Documents\Pythonfordjupning\models_final\G1_RFModel_final.pkl

--- Retraining HGBModel on G1 ---
Saved HGBModel → c:\Users\Eier\Documents\Pythonfordjupning\models_final\G1_HGBModel_final.pkl

--- Retraining CatBoostModel on G1 ---
Saved CatBoostModel → c:\Users\Eier\Documents\Pythonfordjupning\models_final\G1_CatBoostModel_final.pkl

--- Building EnsembleModel for G1 ---
Saved Ensemble model → c:\Users\Eier\Documents\Pythonfordjupning\models_final\G1_Ensemble_final.pkl
```

```
=====
RETRAINING MODELS ON G2
=====

--- Retraining RFModel on G2 ---
Saved RFModel → c:\Users\Eier\Documents\Pythonfordjupning\models_final\G2_RFModel_final.pkl

--- Retraining HGBModel on G2 ---
Saved HGBModel → c:\Users\Eier\Documents\Pythonfordjupning\models_final\G2_HGBModel_final.pkl

--- Retraining CatBoostModel on G2 ---
Saved CatBoostModel → c:\Users\Eier\Documents\Pythonfordjupning\models_final\G2_CatBoostModel_final.pkl

--- Building EnsembleModel for G2 ---
Saved Ensemble model → c:\Users\Eier\Documents\Pythonfordjupning\models_final\G2_Ensemble_final.pkl
```

```

=====
RETRAINING MODELS ON G3
=====

--- Retraining RFModel on G3 ---
Saved RFModel → c:\Users\Eier\Documents\Pythonfordjupning\models_final\G3_RFModel_final.pkl

--- Retraining HGBModel on G3 ---
Saved HGBModel → c:\Users\Eier\Documents\Pythonfordjupning\models_final\G3_HGBModel_final.pkl

--- Retraining CatBoostModel on G3 ---
Saved CatBoostModel → c:\Users\Eier\Documents\Pythonfordjupning\models_final\G3_CatBoostModel_final.pkl

--- Building EnsembleModel for G3 ---
Saved Ensemble model → c:\Users\Eier\Documents\Pythonfordjupning\models_final\G3_Ensemble_final.pkl

```

```

=====
RETRAINING MODELS ON G4
=====

--- Retraining RFModel on G4 ---
Saved RFModel → c:\Users\Eier\Documents\Pythonfordjupning\models_final\G4_RFModel_final.pkl

--- Retraining HGBModel on G4 ---
Saved HGBModel → c:\Users\Eier\Documents\Pythonfordjupning\models_final\G4_HGBModel_final.pkl

--- Retraining CatBoostModel on G4 ---
Saved CatBoostModel → c:\Users\Eier\Documents\Pythonfordjupning\models_final\G4_CatBoostModel_final.pkl

--- Building EnsembleModel for G4 ---
Saved Ensemble model → c:\Users\Eier\Documents\Pythonfordjupning\models_final\G4_Ensemble_final.pkl

=====
RETRAIN PIPELINE COMPLETED
=====

```

Modell körning på testdata set G1- G4 15 %

=== TEST RESULTS (SUMMARY DF) ===

group	model	accuracy	balanced_accuracy
G1	RFModel	0.965438	0.965535
G1	HGBModel	0.963134	0.963390
G1	CatBoostModel	0.957757	0.957759
G1	EnsembleModel	0.969278	0.969378
G2	RFModel	0.961694	0.961977
G2	HGBModel	0.954973	0.955164
G2	CatBoostModel	0.956989	0.957201
G2	EnsembleModel	0.956989	0.957331
G3	RFModel	0.969278	0.968691
G3	HGBModel	0.969278	0.968881
G3	CatBoostModel	0.966974	0.966898
G3	EnsembleModel	0.973118	0.972663
G4	RFModel	0.979564	0.978976
G4	HGBModel	0.967984	0.967107
G4	CatBoostModel	0.976158	0.975523
G4	EnsembleModel	0.978883	0.978327

=== EVALUATION G1 ===

Label distribution (y_test):

y

1 454

0 437

-1 411

Name: count, dtype: int64

Majority class baseline accuracy: 0.3487

--- RFModel (G1) ---

Classification Report:

	precision	recall	f1-score	support
-1	0.98	0.98	0.98	411
0	0.95	0.95	0.95	437
1	0.97	0.97	0.97	454
accuracy		0.97		1302
macro avg	0.97	0.97	0.97	1302
weighted avg	0.97	0.97	0.97	1302

Confusion Matrix:

```
[[401 10  0]
```

```
[ 10 414 13]
```

```
[ 0 12 442]]
```

Feature Importance:

```
[1.18677390e-04 1.01484119e-04 1.21437137e-04 1.27663853e-04
```

```
5.87821742e-04 1.86488000e-03 5.88406095e-03 1.01451728e-04
```

```
1.62414129e-03 5.09708177e-03 1.13855737e-04 2.87392325e-02
```

```
1.10195964e-01 1.23615808e-04 6.04627237e-04 4.73038373e-04
```

```
1.26250528e-03 1.10996913e-03 2.84855826e-03 1.05357495e-04
```

```
1.45455971e-03 5.13730613e-03 1.11337524e-04 5.70080795e-04
```

```
4.80742829e-04 1.25699606e-03 1.46026746e-03 3.86699081e-03
```

```
1.31958856e-04 5.87733909e-04 4.95255473e-04 1.29368670e-03
```

```
5.58500983e-04 4.79459378e-04 1.18999270e-03 3.52944133e-03
```

```
1.18513955e-04 3.12728729e-03 1.07910770e-04 1.57531868e-04
```

--- HGBModel (G1) ---

Classification Report:

	precision	recall	f1-score	support
-1	0.98	0.98	0.98	411
0	0.94	0.95	0.95	437
1	0.97	0.96	0.97	454
accuracy			0.96	1302
macro avg	0.96	0.96	0.96	1302

weighted avg 0.96 0.96 0.96 1302

Confusion Matrix:

[[401 10 0]

[9 416 12]

[0 17 437]]

--- CatBoostModel (G1) ---

Classification Report:

	precision	recall	f1-score	support
-1	0.98	0.95	0.96	411
0	0.91	0.97	0.94	437
1	0.99	0.95	0.97	454

accuracy		0.96	1302
----------	--	------	------

macro avg	0.96	0.96	0.96	1302
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weighted avg	0.96	0.96	0.96	1302
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Confusion Matrix:

[[392 19 0]

[10 423 4]

[0 22 432]]

Feature Importance:

[3.88178423e-02 9.19394538e-02 1.75229558e-02 3.86983318e-02

6.98376325e-02 1.52064129e-01 1.23214082e+00 2.03058072e-03

1.60325871e-01 4.05632556e-01 4.51149658e-04 1.84860481e+00

3.08535692e+00 4.19748312e-02 7.84189372e-02 6.29819043e-02

7.17918824e-02 1.11037633e-01 6.59598738e-01 2.54472711e-02
2.63111145e-01 4.31709095e-01 2.53833830e-04 4.65094792e-02
5.49039653e-02 1.58752710e-01 3.97537707e-01 3.14733621e-01
2.29007071e-02 0.00000000e+00 5.51052737e-02 8.76935209e-02
2.95820237e-02 2.33816642e-02 4.79570064e-02 4.78503252e-01
2.01126121e-02 1.06900372e-01 0.00000000e+00 6.97323745e-02

=====

RUNNING EVALUATION FOR G2

=====

=== EVALUATION G2 ===

Label distribution (y_test):

y

-1 525

0 503

1 460

Name: count, dtype: int64

Majority class baseline accuracy: 0.3528

--- RFModel (G2) ---

Classification Report:

	precision	recall	f1-score	support
-1	0.96	0.96	0.96	525
0	0.94	0.95	0.95	503
1	0.99	0.97	0.98	460

accuracy		0.96	1488	
macro avg	0.96	0.96	0.96	1488
weighted avg	0.96	0.96	0.96	1488

Confusion Matrix:

[[505 20 0]

[18 479 6]

[5 8 447]]

Feature Importance:

[9.80344807e-05 8.06652554e-05 1.08260300e-04 8.09445163e-05
3.71886151e-04 1.05165242e-03 2.41443040e-03 8.22027857e-05
1.37067743e-03 1.50585316e-03 9.22493224e-05 2.08545993e-02
9.74903537e-02 1.11840621e-04 3.87861768e-04 3.83651296e-04
5.85654896e-04 9.21738904e-04 1.92212228e-03 8.97880033e-05
1.30150114e-03 2.47846850e-03 9.29542746e-05 3.57547895e-04
4.65008170e-04 6.69408083e-04 1.23904307e-03 3.63551748e-03
9.82925204e-05 3.55874331e-04 3.82685293e-04 6.00687819e-04
3.53975368e-04 4.00458402e-04 6.06091444e-04 1.38333085e-03
9.79784091e-05 1.27491481e-03 9.29132943e-05 5.53603348e-04

--- HGBModel (G2) ---

Classification Report:

	precision	recall	f1-score	support
-1	0.96	0.96	0.96	525
0	0.93	0.94	0.94	503
1	0.98	0.97	0.97	460

accuracy		0.95		1488
macro avg	0.96	0.96	0.96	1488
weighted avg	0.96	0.95	0.96	1488

Confusion Matrix:

```
[[505 20 0]
```

```
[ 20 472 11]
```

```
[ 3 13 444]]
```

--- CatBoostModel (G2) ---

Classification Report:

	precision	recall	f1-score	support
-1	0.97	0.95	0.96	525
0	0.92	0.95	0.94	503
1	0.98	0.96	0.97	460
accuracy		0.96		1488
macro avg	0.96	0.96	0.96	1488
weighted avg	0.96	0.96	0.96	1488

Confusion Matrix:

```
[[501 24 0]
```

```
[ 13 480 10]
```

```
[ 0 17 443]]
```

Feature Importance:

```
[2.94533529e-04 1.33385773e-02 9.89580815e-03 3.79088743e-03  
3.55880782e-02 8.28714569e-02 1.19866508e-01 1.75624049e-02]
```

1.33130350e-01 2.27148255e-01 4.43874550e-03 5.16089306e-01
2.57204981e+00 0.00000000e+00 3.90016168e-03 1.23557110e-02
4.36494487e-02 1.01443070e-01 4.23318858e-01 4.31873507e-04
1.17305980e-01 1.94343681e-01 7.38519617e-03 2.17516562e-03
2.23547735e-02 3.11748521e-02 2.12213751e-01 3.76558239e-01
8.97084376e-03 1.41113849e-02 3.43718572e-02 2.86574327e-02
1.73732367e-02 1.97922562e-02 3.21489333e-02 1.11271195e-01
2.41735499e-03 1.46329833e-01 1.23905919e-02 8.66386199e-02

=====

RUNNING EVALUATION FOR G3

=====

=== EVALUATION G3 ===

Label distribution (y_test):

y

-1 471

1 429

0 402

Name: count, dtype: int64

Majority class baseline accuracy: 0.3618

--- RFModel (G3) ---

Classification Report:

	precision	recall	f1-score	support
-1	0.98	0.98	0.98	471
0	0.95	0.95	0.95	402

1	0.98	0.98	0.98	429
accuracy			0.97	1302
macro avg	0.97	0.97	0.97	1302
weighted avg	0.97	0.97	0.97	1302

Confusion Matrix:

[[460 11 0]

[11 383 8]

[0 10 419]]

Feature Importance:

[1.33724646e-04 1.31608355e-04 1.56554018e-04 1.46338931e-04
3.31407286e-04 1.32636864e-03 2.54533580e-03 1.12294483e-04
1.19421627e-03 1.51690207e-03 1.39421922e-04 1.51276687e-02
5.00207241e-02 1.17917034e-04 3.28292698e-04 3.54649293e-04
6.56493576e-04 1.23350648e-03 2.09804952e-03 1.23963308e-04
1.44543761e-03 3.30172536e-03 1.53412029e-04 3.26495073e-04
4.01713473e-04 7.03801075e-04 1.39042392e-03 2.81504179e-03
1.28072662e-04 3.58422403e-04 3.99076088e-04 6.86415075e-04
3.19009101e-04 4.47074679e-04 7.21321079e-04 1.52821286e-03
1.28865382e-04 1.42199551e-03 1.22923749e-04 3.15437379e-04

--- HGBModel (G3) ---

Classification Report:

	precision	recall	f1-score	support
-1	0.98	0.97	0.98	471
0	0.95	0.96	0.95	402

1	0.98	0.98	0.98	429
accuracy			0.97	1302
macro avg	0.97	0.97	0.97	1302
weighted avg	0.97	0.97	0.97	1302

Confusion Matrix:

[[458 13 0]

[10 384 8]

[0 9 420]]

--- CatBoostModel (G3) ---

Classification Report:

	precision	recall	f1-score	support
-1	0.98	0.97	0.98	471
0	0.93	0.97	0.95	402
1	0.98	0.97	0.98	429
accuracy			0.97	1302
macro avg	0.97	0.97	0.97	1302
weighted avg	0.97	0.97	0.97	1302

Confusion Matrix:

[[456 15 0]

[7 388 7]

[0 14 415]]

Feature Importance:

[2.11321351e-02 9.12203033e-03 1.69690491e-02 2.11776872e-02
2.27302815e-02 2.19717643e-01 2.62209827e-01 1.31515336e-02

2.03547665e-01 1.88924979e-01 1.03153543e-02 7.43775130e-01
1.48062070e+00 7.37422728e-03 2.59104161e-02 5.54008555e-02
1.02331905e-01 1.92123216e-01 4.39844187e-01 9.43970215e-03
2.19729997e-01 3.95178269e-01 1.54686939e-02 1.47963616e-02
6.56948893e-02 9.60000927e-02 2.10800421e-01 4.20926595e-01
2.43706721e-02 3.60543283e-02 4.88342470e-02 6.92248901e-02
1.26715250e-02 3.72465340e-02 9.62747294e-02 1.73846057e-01
3.88386611e-03 1.79878513e-01 7.57084349e-03 9.49313647e-02

=====

RUNNING EVALUATION FOR G4

=====

=== EVALUATION G4 ===

Label distribution (y_test):

y

-1 514

1 510

0 444

Name: count, dtype: int64

Majority class baseline accuracy: 0.3501

--- RFModel (G4) ---

Classification Report:

	precision	recall	f1-score	support
--	-----------	--------	----------	---------

-1	0.98	0.98	0.98	514
----	------	------	------	-----

0	0.97	0.97	0.97	444
1	0.99	0.99	0.99	510
accuracy		0.98		1468
macro avg	0.98	0.98	0.98	1468
weighted avg	0.98	0.98	0.98	1468

Confusion Matrix:

[[506 8 0]

[9 429 6]

[0 7 503]]

Feature Importance:

[1.50057306e-04 1.41786880e-04 1.36159479e-04 1.55533965e-04
2.99231986e-04 9.37400548e-04 2.64322836e-03 1.39243044e-04
6.76998910e-04 1.38576775e-03 1.35183330e-04 9.65714587e-03
5.80155992e-02 1.32442239e-04 2.96214792e-04 2.65967974e-04
4.47535270e-04 5.61515808e-04 1.09197999e-03 1.37579222e-04
8.95934650e-04 1.79776870e-03 1.52009096e-04 3.27797163e-04
2.71944840e-04 4.65636226e-04 9.21961897e-04 1.60836249e-03
1.60061085e-04 3.13490970e-04 2.67622740e-04 4.50072829e-04
3.23330423e-04 2.26311799e-04 4.56232403e-04 1.97614570e-03
1.37717980e-04 1.23922353e-03 1.43261606e-04 4.50497190e-04

--- HGBModel (G4) ---

Classification Report:

	precision	recall	f1-score	support
--	-----------	--------	----------	---------

-1	0.98	0.98	0.98	514
0	0.95	0.95	0.95	444
1	0.97	0.98	0.97	510
accuracy			0.97	1468
macro avg	0.97	0.97	0.97	1468
weighted avg	0.97	0.97	0.97	1468

Confusion Matrix:

[[502 12 0]

[9 421 14]

[0 12 498]]

--- CatBoostModel (G4) ---

Classification Report:

	precision	recall	f1-score	support
-1	0.99	0.98	0.98	514
0	0.96	0.96	0.96	444
1	0.98	0.98	0.98	510
accuracy			0.98	1468
macro avg	0.98	0.98	0.98	1468
weighted avg	0.98	0.98	0.98	1468

Confusion Matrix:

[[504 10 0]

[7 427 10]

[0 8 502]]

Feature Importance:

[2.34881773e-03 6.92367953e-03 1.53049056e-02 4.98742011e-03
1.45564570e-02 8.41832638e-02 1.33418128e-01 6.75732355e-03
4.89294800e-02 1.80772451e-01 5.35474947e-03 4.37784040e-01
1.76388154e+00 2.51588195e-03 9.24996541e-03 1.07983790e-02
1.32547138e-02 4.02702741e-02 2.47272434e-01 7.83520751e-03
4.74524561e-02 1.59273321e-01 5.99804292e-03 8.68899504e-03
1.03094047e-02 1.36725782e-02 6.57256509e-02 1.75086299e-01
8.16869323e-03 2.35746079e-02 3.50437340e-03 2.52747910e-02
8.78878304e-03 1.29029579e-02 1.05977641e-02 2.28256051e-01
1.07180846e-05 1.71513067e-01 1.59026704e-02 6.30147754e-02]